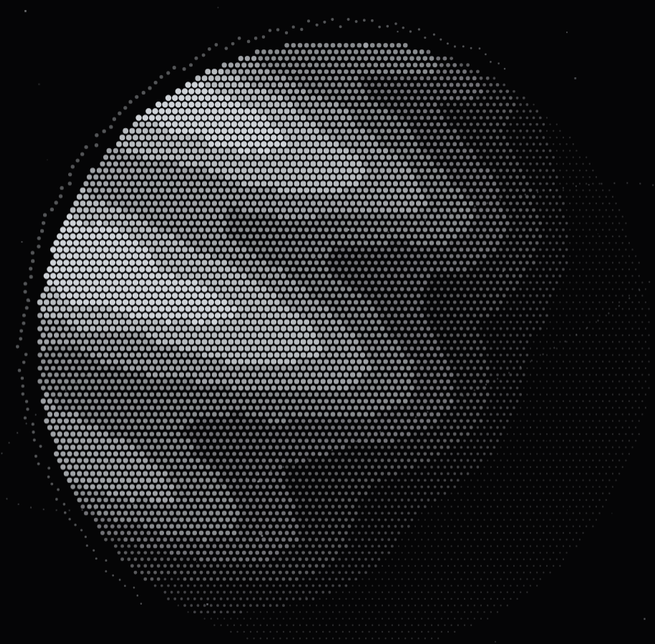
THREE SUB-NEPTUNE PLANET CANDIDATES

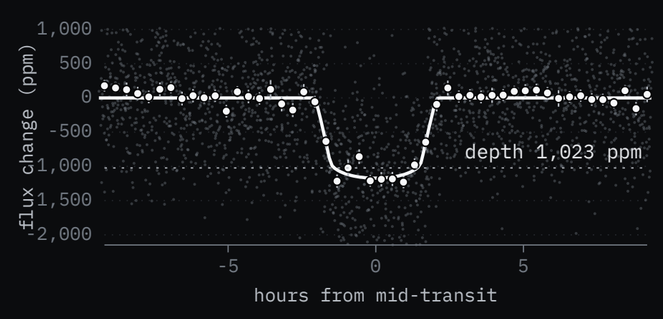


Identified in Kepler data from 2009–2013 around stars the telescope recorded but never targeted. Each passes every test the data allow, with a false-positive probability below 1 %.

THE THREE

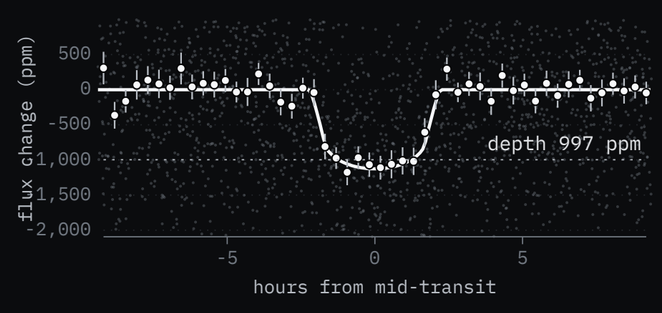
KIC 6699887 — 0.63 %

15.1136-day orbit · 2.76 Earth radii
1,023 ppm · 86 transits · T_{eq} ≈ 611 K
host 4,542 K dwarf, G 16.2, 859 pc
FPP ~~2.4 %~~ → 0.63 % with Gaia limits
second on-target signal at 47.69 d



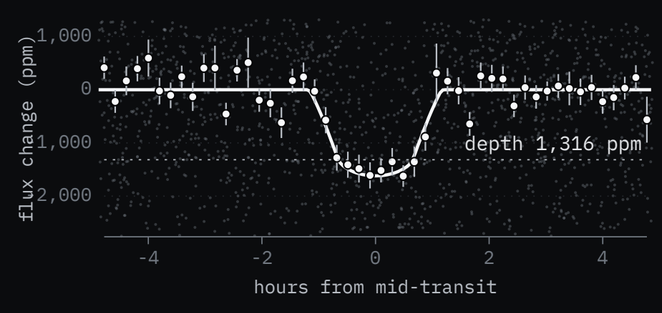
KIC 3831969 — 0.71 %

15.3133-day orbit · 3.47 Earth radii
997 ppm · 86 transits · T_{eq} ≈ 722 K
host 5,275 K dwarf, G 16.5, 1,969 pc
FPP ~~2.7 %~~ → 0.71 % with Gaia limits



KIC 11138423 — 0.70 %

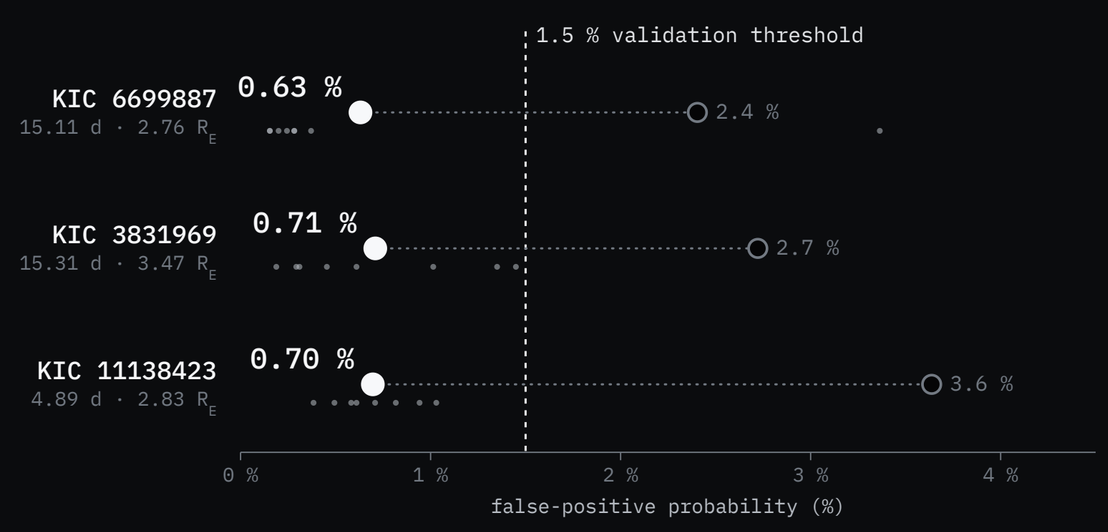
4.8945-day orbit · 2.83 Earth radii
1,316 ppm · 253 transits · T_{eq} ≈ 720 K
host 4,258 K dwarf, G 17.4, 894 pc
FPP ~~3.6 %~~ → 0.70 % with Gaia limits



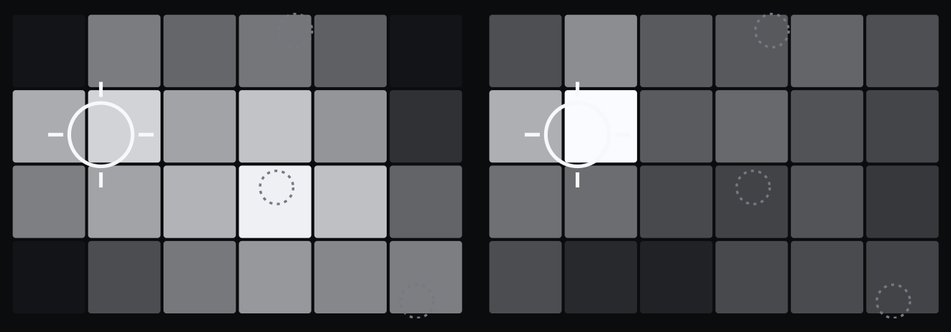
HOW THEY WERE FOUND

6,465	448	22	5	4	**3**
stars searched, thresholds frozen first	first-screen passes (upside-down null: 97)	pass stage-2 vetting	lose light on their own pixels	survive re-derivation and red team	candidates, FPP below 1 %

FALSE-POSITIVE PROBABILITY



THE LIGHT GOES MISSING FROM THE TARGET



KIC 6699887, Kepler season 2. Left: the stamp of the bright star Kepler observed. Right: in-transit minus out-of-transit light. The missing light sits on the candidate's Gaia position (ring): $\Delta\chi^2$ 182 vs no signal, 60 vs the next-best star.

EVERY CONDITION A CANDIDATE MUST MEET

- ✓ **Pre-registered search with a null**
  Frozen target list and thresholds; the same search on every light curve turned upside down.
- ✓ **Light lost on the target's own pixels**
  Two independent pixel tests, per quarter and per season.
- ✓ **Independent re-derivation**
  A separate agent with no shared code recovered period and depth.
- ✓ **Transit shape fits the host star**
  Stellar density from the transit matches Gaia within 0.5 σ.
- ○ **High-resolution imaging**
  Speckle or adaptive optics inside 0.5″ to replace Gaia's statistical limit.
- ✓ **Stage-2 vetting**
  Odd and even transits match; no dip at half phase; every transit consistent; no stellar variability.
- ✓ **No known object shares the ephemeris**
  Checked against every KOI, threshold-crossing event and eclipsing binary in the Kepler field.
- ✓ **Red-team review**
  Agents tasked with killing each signal found no false-positive mechanism.
- ✓ **False-positive probability below 1.5 %**
  TRICERATOPS with Gaia constraints: 0.63, 0.71 and 0.70 %; nearby stars 0 %.
- ○ **Refereed publication**
  The step that turns a candidate into a validated planet.

CANDIDATE

A periodic dip that survives every test the data allow: localised on the star, absent from the inverted-flux null, unmatched by any known object, and unlikely to be anything else.

VALIDATED

False-positive probability below 1.5 % and nearby-star probability below 0.1 % in a refereed paper, usually with a high-resolution image of the host. Not yet done here.

CREDITS

Kepler Bonus photometry (Martínez-Palomera et al. 2023, CC BY 4.0), Gaia DR3, TRICERATOPS, MOLUSC. Search, vetting and figures produced with Claude Code (Anthropic) agents. Illustrations are not to scale.